In [ ]:

```
```

In [1]:

```
pip install pandas seaborn matplotlib gradio ollama
```

```
Requirement already satisfied: pandas in c:\users\ipl4\anaconda3\lib\site-package
s (2.3.1)
Requirement already satisfied: seaborn in c:\users\ipl4\anaconda3\lib\site-packag
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kages (3.10.5)
Collecting gradio
  Downloading gradio-5.44.1-py3-none-any.whl (60.2 MB)
     --------------------------------------- 60.2/60.2 MB 17.2 MB/s eta 0:00:00
Collecting ollama
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Requirement already satisfied: tzdata>=2022.7 in c:\users\ipl4\anaconda3\lib\site
-packages (from pandas) (2025.2)
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e-packages (from matplotlib) (25.0)
Collecting groovy~=0.1
  Downloading groovy-0.1.2-py3-none-any.whl (14 kB)
Collecting gradio-client==1.12.1
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     --------------------------------------- 324.6/324.6 kB ? eta 0:00:00
Collecting ruff>=0.9.3
  Downloading ruff-0.12.11-py3-none-win_amd64.whl (13.0 MB)
     --------------------------------------- 13.0/13.0 MB 31.2 MB/s eta 0:00:00
Collecting ffmpy
  Downloading ffmpy-0.6.1-py3-none-any.whl (5.5 kB)
Collecting orjson~=3.0
  Downloading orjson-3.11.3-cp310-cp310-win_amd64.whl (131 kB)
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b\site-packages (from gradio) (3.0.2)
Collecting brotli>=1.1.0
  Downloading Brotli-1.1.0-cp310-cp310-win_amd64.whl (357 kB)
     --------------------------------------- 357.3/357.3 kB ? eta 0:00:00
Collecting tomlkit<0.14.0,>=0.12.0
  Downloading tomlkit-0.13.3-py3-none-any.whl (38 kB)
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  Downloading aiofiles-24.1.0-py3-none-any.whl (15 kB)
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te-packages (from gradio) (6.0.2)
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\lib\site-packages (from gradio) (4.12.2)
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kages (from gradio) (3.1.6)
Collecting pydub
  Downloading pydub-0.25.1-py2.py3-none-any.whl (32 kB)
Collecting uvicorn>=0.14.0
  Downloading uvicorn-0.35.0-py3-none-any.whl (66 kB)
     ---------------------------------------- 66.4/66.4 kB ? eta 0:00:00
Collecting semantic-version~=2.0
  Downloading semantic_version-2.10.0-py2.py3-none-any.whl (15 kB)
Requirement already satisfied: pydantic<2.12,>=2.0 in c:\users\ipl4\anaconda3\lib
\site-packages (from gradio) (2.11.7)
Requirement already satisfied: httpx<1.0,>=0.24.1 in c:\users\ipl4\anaconda3\lib
\site-packages (from gradio) (0.28.1)
Collecting huggingface-hub<1.0,>=0.33.5
  Downloading huggingface_hub-0.34.4-py3-none-any.whl (561 kB)
     ------------------------------------- 561.5/561.5 kB 36.7 MB/s eta 0:00:00
Requirement already satisfied: anyio<5.0,>=3.0 in c:\users\ipl4\anaconda3\lib\sit
e-packages (from gradio) (4.7.0)
Collecting starlette<1.0,>=0.40.0
  Downloading starlette-0.47.3-py3-none-any.whl (72 kB)
     ---------------------------------------- 73.0/73.0 kB 3.9 MB/s eta 0:00:00
Collecting safehttpx<0.2.0,>=0.1.6
  Downloading safehttpx-0.1.6-py3-none-any.whl (8.7 kB)
Collecting fastapi<1.0,>=0.115.2
  Downloading fastapi-0.116.1-py3-none-any.whl (95 kB)
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Collecting typer<1.0,>=0.12
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Collecting python-multipart>=0.0.18
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Collecting fsspec
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Collecting websockets<16.0,>=10.0
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Requirement already satisfied: idna>=2.8 in c:\users\ipl4\anaconda3\lib\site-pack
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Requirement already satisfied: certifi in c:\users\ipl4\anaconda3\lib\site-packag
es (from httpx<1.0,>=0.24.1->gradio) (2025.7.14)
Requirement already satisfied: h11>=0.16 in c:\users\ipl4\anaconda3\lib\site-pack
ages (from httpcore==1.*->httpx<1.0,>=0.24.1->gradio) (0.16.0)
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ges (from huggingface-hub<1.0,>=0.33.5->gradio) (2.32.4)
Requirement already satisfied: tqdm>=4.42.1 in c:\users\ipl4\anaconda3\lib\site-p
ackages (from huggingface-hub<1.0,>=0.33.5->gradio) (4.67.1)
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ges (from huggingface-hub<1.0,>=0.33.5->gradio) (3.17.0)
Requirement already satisfied: typing-inspection>=0.4.0 in c:\users\ipl4\anaconda
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Requirement already satisfied: annotated-types>=0.6.0 in c:\users\ipl4\anaconda3
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Requirement already satisfied: six>=1.5 in c:\users\ipl4\anaconda3\lib\site-packa
ges (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: rich>=10.11.0 in c:\users\ipl4\anaconda3\lib\site-
packages (from typer<1.0,>=0.12->gradio) (13.9.4)
Requirement already satisfied: shellingham>=1.3.0 in c:\users\ipl4\anaconda3\lib
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Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\ipl4\anaconda3
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Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\ipl4\anaconda3\l
ib\site-packages (from rich>=10.11.0->typer<1.0,>=0.12->gradio) (2.2.0)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\ipl4\anaconda
3\lib\site-packages (from requests->huggingface-hub<1.0,>=0.33.5->gradio) (3.3.2)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\ipl4\anaconda3\lib
\site-packages (from requests->huggingface-hub<1.0,>=0.33.5->gradio) (2.5.0)
Requirement already satisfied: mdurl~=0.1 in c:\users\ipl4\anaconda3\lib\site-pac
kages (from markdown-it-py>=2.2.0->rich>=10.11.0->typer<1.0,>=0.12->gradio) (0.1.
0)
Installing collected packages: pydub, brotli, websockets, tomlkit, semantic-versi
on, ruff, python-multipart, orjson, groovy, fsspec, ffmpy, aiofiles, uvicorn, sta
rlette, huggingface-hub, typer, safehttpx, ollama, gradio-client, fastapi, gradio
  Attempting uninstall: typer
    Found existing installation: typer 0.9.0
    Uninstalling typer-0.9.0:
      Successfully uninstalled typer-0.9.0
Successfully installed aiofiles-24.1.0 brotli-1.1.0 fastapi-0.116.1 ffmpy-0.6.1 f
sspec-2025.9.0 gradio-5.44.1 gradio-client-1.12.1 groovy-0.1.2 huggingface-hub-0.
34.4 ollama-0.5.3 orjson-3.11.3 pydub-0.25.1 python-multipart-0.0.20 ruff-0.12.11
safehttpx-0.1.6 semantic-version-2.10.0 starlette-0.47.3 tomlkit-0.13.3 typer-0.1
7.3 uvicorn-0.35.0 websockets-15.0.1
Note: you may need to restart the kernel to use updated packages.
```

In [2]:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load Titanic Dataset
url = r"C:\Users\IPL4\Downloads\titanic_ dataset_final.csv"
df = pd.read_csv(url)
df
```

Out[2]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450
...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376

891 rows × 12 columns

In [3]:

```
# Display dataset info
print(df.describe())
```

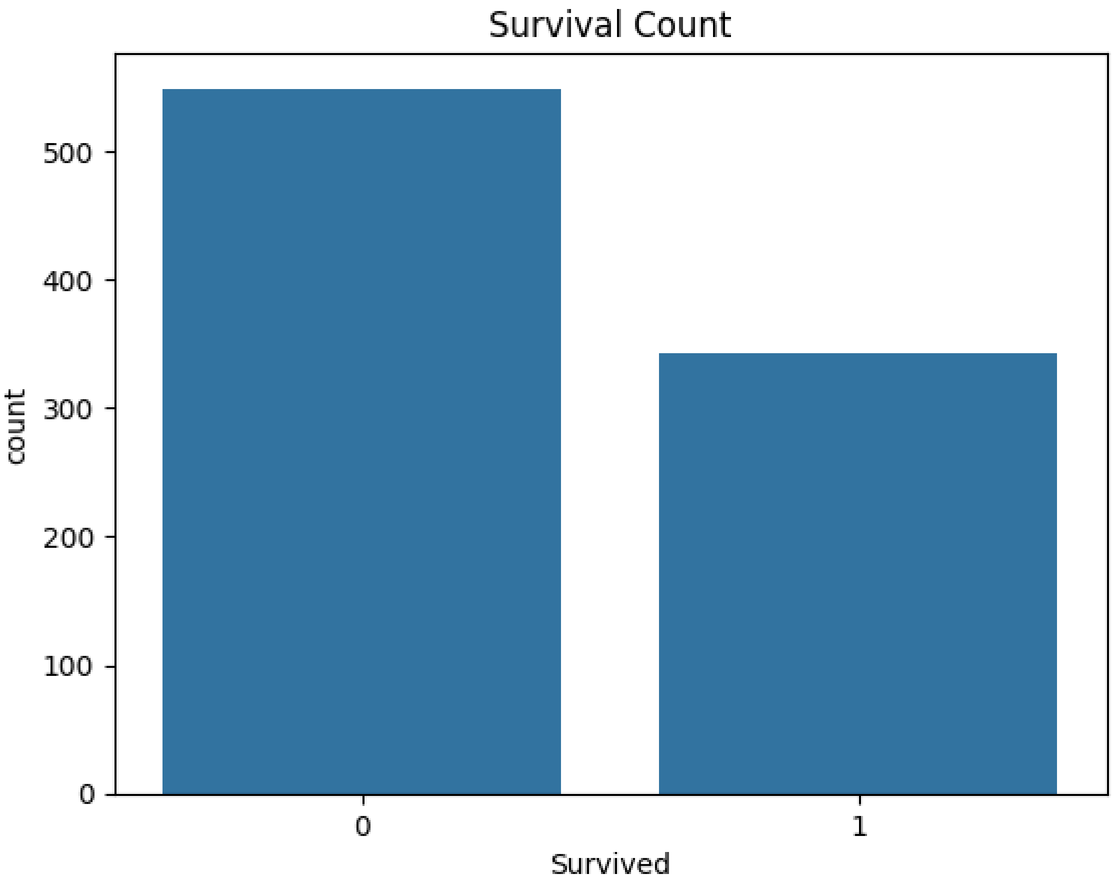
```
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

In [4]:
```python
# Missing Values Check
print("\nMissing Values:\n", df.isnull().sum())
```

```
Missing Values:
 PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [5]:
```python
# Survival Rate Visualization
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()
```



In [6]:
```
import ollama

def generate_insights(df_summary):
    prompt = f"Analyze the dataset summary and provide insights:\n\n{df_summary}
    response = ollama.chat(model="gemma:2b", messages=[{"role": "user", "content
    return response['message']['content']

# Generate AI Insights
summary = df.describe().to_string()
insights = generate_insights(summary)
print("\n◆ AI-Generated Insights:\n", insights)
```

```
🔷 AI-Generated Insights:
 ## Insights from the dataset summary:

**Overall:**

* The dataset contains information about passengers who survived a plane crash in
different classes.
* The data covers a wide range of characteristics, including age, gender, family
situation, and travel details.

**Key features:**

* **Age:** The average age of passengers who survived is 30.7 years old.
* **Gender:** The data is balanced, with an equal number of male and female passe
ngers.
* **Family situation:** Most passengers were married (84%).
* **Travel class:** The majority of passengers traveled in class 3 (34%).

**Additional insights:**

* The minimum and maximum values for several variables indicate that there are pa
ssengers with very different experiences.
* The presence of a large number of missing values (20.2% of the dataset) could p
otentially affect the analysis.
* The high number of missing values in the fare variable might be due to the dive
rse range of travel classes and the lack of information about the cost of transpo
rtation.

**Further analysis:**

* Analyzing the distribution of continuous variables like age, gender, and family
situation.
* Exploring the relationships between different variables.
* Identifying potential outliers and investigating why they might have experience
d the crash.
* Analyzing the data for different groups defined by the "pclass" variable.

**Overall, the dataset provides valuable insights into the demographics and trave
l characteristics of passengers who survived a plane crash.**
```

In [7]:

```python
import gradio as gr

def eda_analysis(file):
    df = pd.read_csv(file.name)
    summary = df.describe().to_string()
    insights = generate_insights(summary)
    return insights

# Create Web Interface
demo = gr.Interface(fn=eda_analysis, inputs="file", outputs="text", title="AI-Pc

# Launch App
demo.launch(share=True)  # Use share=True for Google Colab
```

```
C:\Users\IPL4\anaconda3\lib\site-packages\tqdm\auto.py:21: TqdmWarning: IProgress
not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedo
cs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
```

```
* Running on local URL:  http://127.0.0.1:7861
* Running on public URL: https://a28705654ba8aa2166.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU upgrades, r
un `gradio deploy` from the terminal in the working directory to deploy to Huggin
g Face Spaces (https://huggingface.co/spaces)
```

Out[7]: